

# RacquetAI — Build Day Report

August 15–16, 2026 · Prepared by **Claude** (Anthropic Claude Code — models Claude Fable 5 & Claude Opus 5), operating under the direction of Karan.

Repository: [github.com/Karanvir1729/RacquetAI](https://github.com/Karanvir1729/RacquetAI) · Open pull request: [#1](#) — [squash-head teardrop brand mark](#)

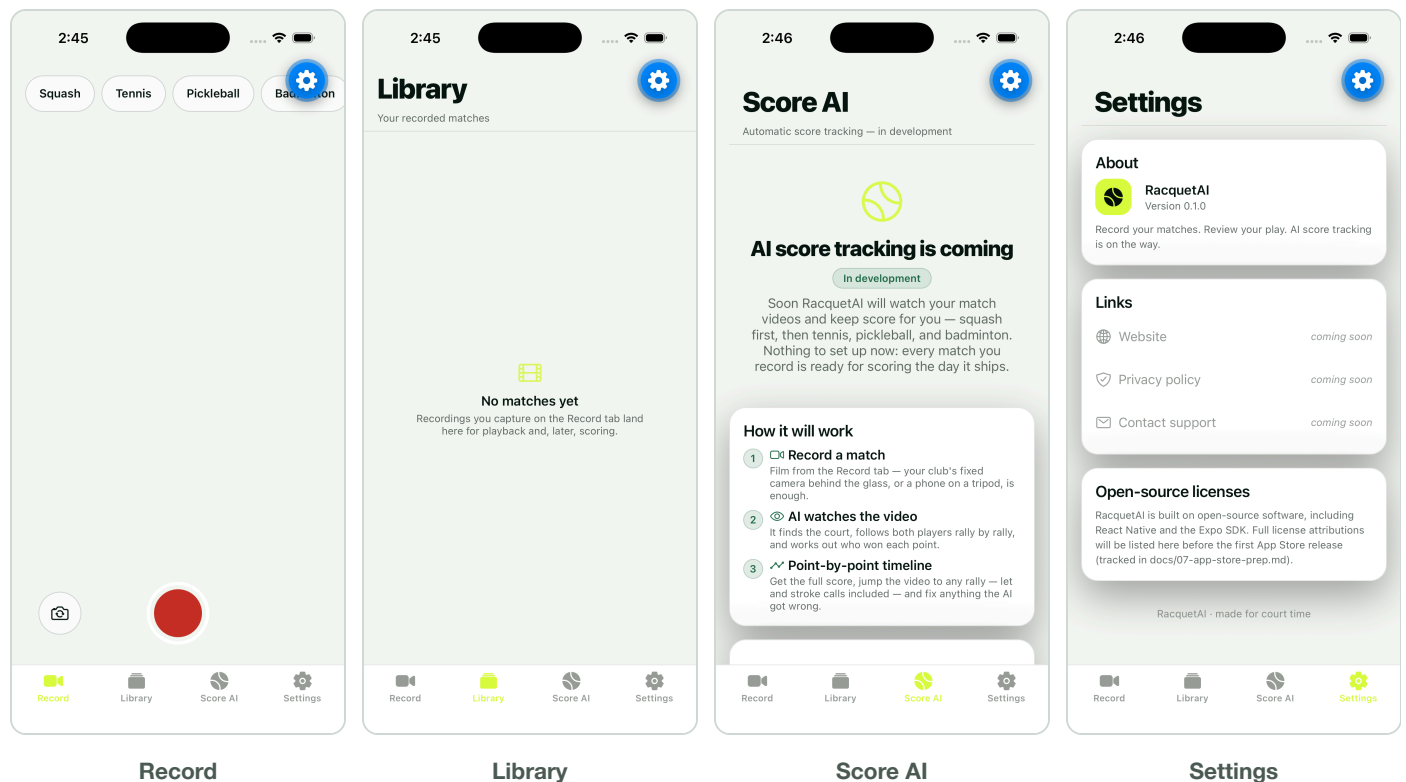
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## Executive summary

In one evening (6:54 PM – 10:33 PM EDT, Aug 15), Claude took RacquetAI from an empty directory to a **public GitHub repository containing a working, tested iOS app** — then verified it live in the iOS Simulator and produced this report the same night.

- **18 commits on `main`** (79 files, ~38,000 insertions), plus 4 feature branches and 1 open PR — all pushed to GitHub.
- **A working Expo/React Native iOS app** (v0.1.0): match video recording, a video library with playback, real squash (PAR-11) and tennis scoring engines behind an honest “in development” Score AI screen, and a settings screen — verified running in the iOS Simulator (screenshots below).
- **Quality gates green:** 178/178 Jest tests passing, TypeScript strict typecheck clean, lint clean, secrets audit clean (no credentials inherited from the predecessor app).
- **Complete brand identity** (“Ace Spark” / “Sweet Spot” icon) with an SVG-master → PNG asset pipeline, plus a competing squash-specific mark now open as PR #1.
- **Full documentation set:** architecture, 5 ADRs, testing policy, App Store submission runbook, an AI-scoring build plan scoped to a US\$10K Azure budget, and a 7-day execution plan delivered as an Excalidraw board.
- **Token spend:** ~302.5 million tokens processed / ~2.49 million generated across 4 Claude sessions and 36 subagents — an estimated **US\$424–498 at Anthropic API list rates**. Full breakdown in the companion *Token Usage & Cost Report*.

## The app, running (iOS Simulator, iPhone 17 — captured Aug 16)



What the screenshots show: the sport selector (Squash first, then Tennis / Pickleball / Badminton), the record control with camera-flip, the library's empty state, the deliberately honest "AI score tracking is coming" screen with the three-step product explanation, and Settings displaying the brand icon at v0.1.0. The blue gear in the corner is the Expo Go development overlay, not part of the app.

## What shipped

### Features (with code locations)

- **Video recording** — `src/features/recording/`: `RecordScreen`, `RecordControls`, `PermissionGate` (graceful camera-permission handling), `SportChips`; built on `expo-camera` per ADR-003.
- **Video library + playback** — sidecar metadata storage ( `storage.ts` , `metadata.ts` , `naming.ts` ), `LibraryScreen`, `PlayerModal`, with unit tests.
- **Scoring engines** — `src/features/scoring/`: a shared `ScoreEngine` interface with a **squash PAR-11 engine** (lets/strokes/no-lets as first-class events) and a tennis state machine, fully unit-tested. The UI ships an honest "coming soon" screen instead of fake AI.
- **App shell** — 4-tab `expo-router` layout, shared component library, token-only dark/light theming, iPad layout support from day one (ADR-002 — the predecessor app was once rejected on iPad grounds).

## Brand



“Ace Spark” identity: 5 SVG masters, a 9-colour palette with measured WCAG contrast ratios, and a `sharp`-based pipeline that regenerates and *verifies* every PNG (icon alpha rules, Android safe zones, silhouette purity). The **“Sweet Spot” app icon** (left) won a three-way parallel design competition judged at 1024/120/48 px. A competing **squash-accurate teardrop mark**, built in an isolated worktree, is open for review as [PR #1](#) — management can pick a direction.

## Documentation & planning

- `docs/01-architecture.md` — app structure and coding rules.
- `docs/02-daybot-infra-heritage.md` — what was inherited from the operator’s prior App-Store-shipped app (infrastructure only; no accounts, secrets, or brand).
- `docs/03-ai-scoring-plan.md` — the AI build contract: squash-first, US\$10K Azure budget with per-phase allocations and kill criteria.
- `docs/06-decisions.md` — 5 ADRs, including ADR-005: a real Expo Go crash (worklets/reanimated native mismatch) diagnosed from the crash report and fixed the same evening.
- `docs/07-app-store-prep.md` — submission runbook written from the predecessor’s actual rejection experience.
- **Week-1 plan** (`docs/planning/`) — a 7-day execution board (Aug 15–21) ending with a real club recording a box-league match off TestFlight, delivered onto the user’s live `excalidraw.com` board and reproducible from source:

[illegible]

## How the day unfolded

| Time (EDT)       | What happened                                                                                                                                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6:54 PM          | Karan asked Claude for a new repo copying the proven iOS infrastructure of his shipped app (read-only reference), with video recording, honestly-stubbed AI scoring plus a real implementation plan, and full branding — built in parallel worktrees. |
| 7:00 PM          | Repo created locally and on GitHub (private). Claude launched a 7-agent workflow: scaffold → 4 parallel feature worktrees → integrator → fresh-eyes auditor.                                                                                          |
| ~7:10 PM         | Session forked. The fork resumed the build workflow after editing it to make <b>squash the primary sport</b> , then ran three more workflows to author, adversarially critique, and render the week-1 Excalidraw board.                               |
| 7:40 PM          | All four feature branches merged to <code>main</code> with zero manual conflicts; audit committed fixes; everything pushed to GitHub.                                                                                                                 |
| 7:52–8:00 PM     | First simulator run crashed Expo Go three times. Claude read the crash report, diagnosed a native-module version mismatch, realigned the dependency matrix (ADR-005), re-ran all tests, and hand-tested all four tabs in light and dark mode.         |
| 8:28 PM          | Week-1 board committed and injected into Karan’s live excalidraw.com tab (after Claude caught and owned a mix-up: <i>“I verified the wrong window.”</i> ).                                                                                            |
| 9:38 PM          | Karan: <i>“squash first is right, make it public.”</i> Repo flipped public; a 3-designer icon competition launched.                                                                                                                                   |
| 10:06–10:23 PM   | A dedicated worktree session (Opus 5) redrew the brand mark with true squash teardrop geometry — now <b>PR #1</b> .                                                                                                                                   |
| 10:32 PM         | Winning “Sweet Spot” icon integrated, verified, committed, pushed. Build day ends.                                                                                                                                                                    |
| 2:38 AM (Aug 16) | This reporting session: pushed the last pending branch, opened PR #1, re-ran the app in the simulator for the screenshots above, mined all session transcripts, and produced these reports.                                                           |

## How it was built: multi-agent orchestration

- This was not one chat thread. Claude coordinated **4 sessions and 36 subagents** across 5 multi-agent workflows:
- **Build workflow (7 agents):** one scaffolder, four parallel feature builders in isolated git worktrees (video, scoring, branding, infra-docs), one integrator, one fresh-eyes security auditor. Merged with zero manual conflicts.
  - **Board workflows:** parallel authoring of the 7-day plan and squash-club product definition, a deterministic Python renderer with overlap self-checks, and a separate **adversarial critique pass** that cut real risks (dropped the Score AI tab from v0.1.0 as an App Review risk, moved the iPad pass before TestFlight, made Azure CPU-only for v0).

- **Icon competition (3 agents):** three designers iterated and self-critiqued renders at 1024/120/48 px; the winner was judged on home-screen legibility.
- **Report workflow (4 agents, this session):** mined 40+ MB of session transcripts to reconstruct this narrative with evidence.

## Verification (evidence, not claims)

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- `npm test` — **178/178 tests passing** (re-run during report preparation, Aug 16).
- `npm run typecheck` — TypeScript strict, zero errors.
- Secrets audit — zero predecessor credentials or identifiers in the tree or git history (verified *before* the repo was made public).
- Live simulator pass — all four tabs exercised, light and dark mode, camera-permission edge case included; re-verified for this report.

## Open items for management

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1. **Pick a brand direction** — “Sweet Spot” Optic-head icon (on `main`) vs. the squash-accurate teardrop mark ([PR #1](#)).
2. **Week-1 plan is live** — 5 tracks targeting a real club match recorded off TestFlight by Aug 21; Day-1 outreach to a known club.
3. **AI scoring** — engines and plan exist; model work is budget-gated (US\$10K Azure, per-phase kill criteria) and intentionally not faked in the UI.